Pneumonia Detection and Explainability Using Vision Transformers and Large Language Models

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***Abstract*—Pneumonia is a solemn universal health issue that mainly disturbs the old and young. Despite being a common diagnostic system, chest X-rays can be problematic in deciding between healthy and pneumonia-affected lungs due to their slight variations. The goal of this project is to advance pneumonia detection and interpretability by applying cutting-edge machine learning methods, particularly Large Language Models (LLMs) and Vision Transformers (ViTs).**

Using the Chest X-Ray Images (Pneumonia) dataset from Kaggle, which consists of 5,863 labeled images, we preprocess the data using augmentation, normalization, and scaling methods. X-rays are processed in patches by the ViT model, which detects important lung areas suggestive of pneumonia and records long-range dependencies. The model's output include classification findings, attention maps that emphasize important patches, and contextually relevant images that are related and retrieved using k-nearest Neighbors (k-NN).

A pre-trained LLM bridges the gap between sophisticated AI processes and the interpretative demands of healthcare practitioners by producing human-readable explanations based on the model's outputs to improve explainability. To guarantee clinical application, criteria such as accuracy, precision, recall, and F1-score are used to assess the suggested system's performance.

The research involves putting together the high-level natural language processing skills of LLMs with the state-of-the-art image analysis capabilities of ViTs to create an effective diagnostic tool that is easy to understand. By improving the diagnoses and availing quick valuable recommendations for medical imaging and artificial intelligence-based health care solutions, this study attempts to bridge a gap.

I. Introduction

Pneumonia is among the primary causes of death worldwide, especially among those who do not have strong immune systems such as kids, old men and women as well as those individuals with decreased immunity. The key to effective cure for and control of pneumonia lies in making sure that it is diagnosed early and accurately. The most common diagnostic method for identifying pneumonia is a chest X-ray (CXR), although interpreting these images can be subjective and frequently calls for skilled radiologists. This emphasizes the necessity of automated, precise, and comprehensible diagnostic tools that can support medical practitioners.

Machine learning models have demonstrated great promise in medical image analysis because of advances in artificial intelligence (AI). Because they can analyze images using a unique patch-based method and capture long-range dependencies, Vision Transformers (ViTs) have become a potent tool for image classification. Adoption of AI-driven solutions in healthcare settings is still hampered by their inability to be interpreted.

This study investigates the use of Large Language Models (LLMs) to improve interpretability in conjunction with Vision Transformers for automated pneumonia identification in chest X-rays. After processing X-ray pictures, the ViT model accurately classifies them by identifying important regions linked to pneumonia. By offering natural language explanations based on model outputs, LLMs help doctors overcome the interpretability problem and make the diagnostic process more transparent and useful.

The project uses intensive training, explainability mechanisms, and state-of-the-art preprocessing methods on the Kaggle Chest X-Ray Images (Pneumonia) dataset. To guarantee dependability and robustness, performance is assessed using metrics such as accuracy, precision, recall, and F1-score. This study intends to improve the diagnostic accuracy and interpretability of pneumonia detection by fusing state-of-the-art machine learning with domain-specific insights, thereby adding to AI's revolutionary influence in healthcare.

II. Review of Literature

Due to the prevalence of pneumonia worldwide as well as developments in medical imaging and artificial intelligence, the creation of automated methods for detecting pneumonia has become a major topic of study. This section examines the development of approaches, and the most recent strategies used to address this problem.

A. Advancing Lung Cancer Detection using Vision Transformers: A Review

Li, A., et al. (2023) reviewed the potential of Vision Transformers (ViTs) in medical imaging, demonstrating that ViTs

outperform Convolutional Neural Networks (CNNs) by capturing global image dependencies. This characteristic makes ViTs particularly suitable for complex medical image analysis tasks, such as chest X-rays, where global patterns play a crucial role in diagnosis.

B. CNNs versus Vision Transformers in Lung Cancer Detection: A Comparative Study

Gai, J., et al. (2023) conducted a comparative study between CNNs and Vision Transformers for lung cancer detection. Their findings indicate that Vision Transformers outperform CNNs, especially when utilizing pre-trained models and large datasets. This emphasizes the importance of leveraging transformer-based architectures for large-scale medical imaging tasks.

C. Efficient Pneumonia Detection using Vision Transformers on Chest X-rays

Singh, S., et al. (2024) explored the application of Vision Transformers for pneumonia detection using chest X-rays. The study highlights that Vision Transformers are more effective than CNNs, particularly when trained on large datasets with data augmentation techniques. This underscores the effectiveness of transformers in improving diagnostic accuracy in medical imaging.

D. Explainable Deep Learning Diagnostic System for Prediction of Lung Disease from Medical Images

Hroub, N. A., et al. (2024) introduced explainable AI techniques, such as Grad-CAM, for visualizing and understanding the decision-making processes of deep learning models. Their work focuses on predicting diseases like pneumonia and COVID-19 from X-rays, ensuring that the model's decisions are interpretable and trustworthy for medical practitioners.

E. Weakly Supervised One-Stage Vision and Language Disease Detection

Tam, L. K., et al. (2020) combined Vision Transformers (ViTs) with Large Language Models (LLMs) for disease detection. The integration of visual data (X-rays) and textual data (patient history) improved diagnostic performance, showcasing the potential of multi-modal approaches in medical imaging and disease detection.

III. Objectives of the Study

The fundamental purpose of this work is to create and deploy an advanced artificial intelligence (AI)-based diagnostic system that can accurately and interpretably diagnose pneumonia from chest X-ray pictures. Pneumonia, a severe respiratory infection, is still a primary source of morbidity and mortality globally, especially in areas with limited access to competent radiologists and healthcare facilities. This research attempts to solve these difficulties by using state-of-the-art AI technologies, notably Vision Transformers (ViTs) and Large Language Models (LLMs), to bridge the gap between medical imaging demands and technology breakthroughs.

At the center of this research is the creation of a Vision Transformer model, which has been tuned for the binary categorization of chest X-rays into two categories: normal and pneumonia. Vision Transformers are preferred over typical convolutional neural networks (CNNs) because to their improved capacity to record global connections within an image, which is critical for detecting subtle patterns in medical data. Traditional approaches frequently fail to identify these patterns, which include lung opacity, consolidation, and textural abnormalities. Using ViTs, the project aims to enhance diagnosis accuracy and assure generalizability across different and heterogeneous medical imaging datasets.

In addition to strong performance, the study underlines the need of interpretability, which is crucial in the healthcare arena where AI systems must instill trust and transparency in medical practitioners. To do this, attention mechanisms are incorporated into the model, resulting in visual explanations known as attention maps. These maps illustrate the particular locations of the X-rays that affected the model's predictions, making the decision-making process more apparent and helping physicians validate the results.

The study's other main goal is to improve interpretability even more by using an LLM to produce written justifications for the model's predictions. By describing the patterns seen in the X-rays and providing explanations for the categorization outcomes, this feature seeks to offer useful insights. By extracting and referring to the most visually comparable cases from the dataset, the LLM also integrates contextual information, enabling medical practitioners to evaluate the model's predictions in a more comprehensive diagnostic context.

Last but not least, the study intends to present its findings via an intuitive application developed using Streamlit to guarantee practical usability. Uploading chest X-rays, getting real-time forecasts, seeing attention maps, looking at similar instances, and accessing thorough written explanations are all features of the tool. The application meets the needs of physicians by emphasizing usability and accessibility, especially in settings with limited resources. This thorough approach highlights the study's overarching goal of enhancing pneumonia diagnosis through the integration of clinical relevance and technology innovation.

IV. Data Collection

The Chest X-Ray Images (Pneumonia) dataset, a publicly accessible resource housed on Kaggle, is the basis for this work. This dataset, which offers a varied collection of chest X-rays classified into two different categories—Normal and Pneumonia—was chosen because it was appropriate for the study's goals. Bacterial and viral cases are also included in the pneumonia category, which makes the dataset complete and representative of actual diagnostic difficulties. The dataset, which consists of more than 5,800 photos with different quality and resolutions, provides a strong basis for the Vision Transformer model's testing, validation, and training.

Three subsets of the dataset are organized into it, and each has a specific function in the modeling process:

1. Training Set: By exposing the Vision Transformer model to a broad variety of patterns and variances in X-ray imagery, this subset—which includes the bulk of the images—is utilized to train the model.

2. Validation Set: A distinct subset is used for hyperparameter adjustment and performance tracking of the model throughout training. This phase prevents overfitting and guarantees that the model generalizes well.

3. Test Set: This subset is used to gauge the model's efficacy and accuracy on unknown data and is only kept for the final assessment.

With a disproportionately large number of pneumonia cases relative to normal patients, class imbalance was a significant concern in the dataset. A biased model that overpredicts the majority class could result from this imbalance if it is left unchecked. Several tactics were put into place to lessen this. The training data underwent considerable use of data augmentation techniques, including flipping, random rotation, brightness modifications, and contrast enhancements. In addition to broadening the dataset's diversity, these changes strengthened the model's resistance to changes in image quality. To further guarantee equal learning outcomes, the minority class was oversampled and weighted loss functions were used during training.

Another essential step in getting the dataset ready for analysis was preprocessing. Every image was downsized to 224 by 224 pixels in order to comply with the Vision Transformer model's input specifications. The patch-based tokenization process at the heart of the ViT architecture, which divides images into smaller pieces for examination, is made easier by this scaling. In order to guarantee numerical consistency throughout the dataset, pixel values were normalized to a defined range. Crucially, medical X-rays' single-channel nature was maintained, preserving the grayscale images' integrity while modifying them for AI processing.

To encourage openness and reproducibility, the trained model checkpoint (`model_state_224.pth`) and the Streamlit application script (`app.py`) are additional files that are essential to the study's execution in addition to the dataset. These tools were crucial to the creation and implementation of the pneumonia detection system, guaranteeing that the larger medical and research community could easily access and apply the study's conclusions.

By combining meticulous preprocessing and augmentation methods with a carefully selected dataset, this study establishes a solid basis for accomplishing its goals. While resources are made publicly available to help future study and application in the field of AI-driven medical diagnostics, careful data processing guarantees the accuracy of the results.

V. Exploratory Data Analysis (EDA) and Hypotheses for the Study

The data set used for the research has around 6000 chest x-ray images with the training set consisting of around 5000 images and the test data set consisting of about 600 images. The images are classified in a binary manner with the class labeled as either normal or pneumonia. The EDA revealed that images are in different sizes and black, white images.

Presence of noticeable class imbalance between the normal and pneumonia cases This imbalance necessitates the implementation of robust preprocessing techniques, such as data augmentation. The preprocessing steps applied to images include multistep transformation, aiming to enhance the feature extraction capability of vision transform model.

Based on the data set composition and the insights gained from the EDA. Several hypotheses can be formulated to get the study. The primary hypothesis is the vision Transformer model will be able to accurately detect and classify pneumonia cases from the images, we can use the model performance on the test set for this hypothesis. Additionally, the study will investigate the interpretability of the model's predictions, Exploring the relationship between the important patches, vector representation and model's classification outputs. The Null Hypothesis: The text generated by the LLM integration is not in the context with insights and prediction derived from the model.

The Alternative Hypothesis: The LLM Genres provides a human readable explanation that is aligned with and contextually related to the vision transformer models outputs.

VI. Data Analytics and Data Visualization

The overall architecture of the data analytics process for the pneumonia detection task. This methodology can be broken down into several key steps , each of which play a crucial role in the overall analysis The process begins with the

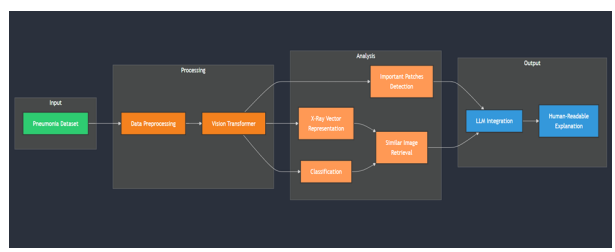


Fig. 1. Architecture

pneumonia data set, which serves as the primary input for the analysis. The input data is undergoing a series of preprocessing steps to prepare it for subsequent analysis. The preprocessed image are divided into patches. The patches are then passed through vision transformer at deep learning architecture that has gained significant attention in the field of computer vision. The Washington transformer takes the input image and applies a series of attention-based ransoms to extract meaningful features from the data. The vision transformer model generates a vector representation of the input X ray images capturing the essential characteristics and patterns within the data. This vector representation typically is of 768 dimensions serves as compact and informative encoding of the input images. The vision transformer also identifies the most important batches of the reasons within the input image that are crucial for classification task. This information can be used to highlight

the specific areas of the X ray images that are most relevant for detection of pneumonia. After classification, the system retrieval of similar X-Ray measures from the data set by comparing the vector representation of the input to those of the images and the data set. The system can identify the most visually similar X ray images which can be useful for further analysis.

A. Data Augmentation

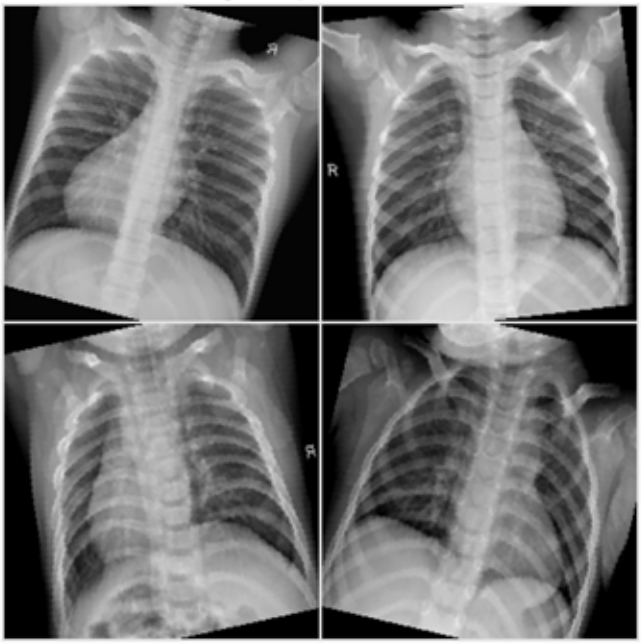
Data augmentation is a crucial step in the data preprocessing stage as it significantly improves the performance of the Vision Transformer model and enhances generalization. The training data undergoes several transformations to expand the diversity of the dataset and make the model more robust to variations in the input.

All the input images are resized to uniform dimensions of 224×224. This resolution is widely adopted for compatibility with deep learning architectures. The transformation *Grayscale* converts the RGB images into grayscale, reducing data dimensionality while emphasizing relevant structural features critical for analysis. This step potentially improves the model's ability to focus on the relevant radiographic features.

The *RandomRotation* and *RandomHorizontalFlip* augmentations introduce controlled variations by rotating and flipping images, making the model more invariant to these transformations. Additionally, the *ColorJitter* transformation randomly adjusts the brightness and contrast of the images, further increasing the diversity of the training data.

By applying these data augmentation methods, the model is exposed to a wide range of variations in the input data, which improves generalization and enhances performance on both training and test sets.

Image examples from dataset



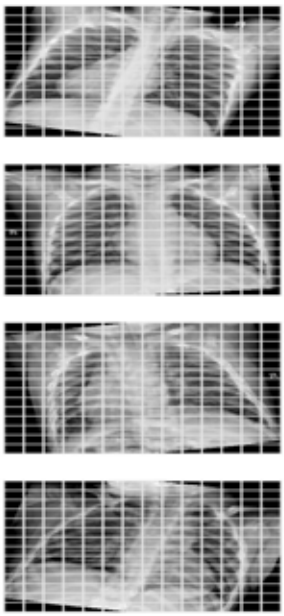
B. Vision Transformer

The Vision Transformer (ViT) model used in this pneumonia detection study demonstrates a multifactor approach to medical image classification, producing three critical outputs: image classification, embedded vector representation, and attention maps.

The input images, after augmentation, are processed through a patch extraction function, which divides the images into a grid of smaller patches of size 16×16. This results in a total of 196 patches per image, where each patch represents a localized region of the original image.

The extracted patches are then linearly projected into a higher-dimensional embedding space using a fully connected layer. These embedded representations are added with positional encoding vectors, which allow the Vision Transformer to access positional information about the patches. Additionally, a class token is introduced to capture global, image-level features.

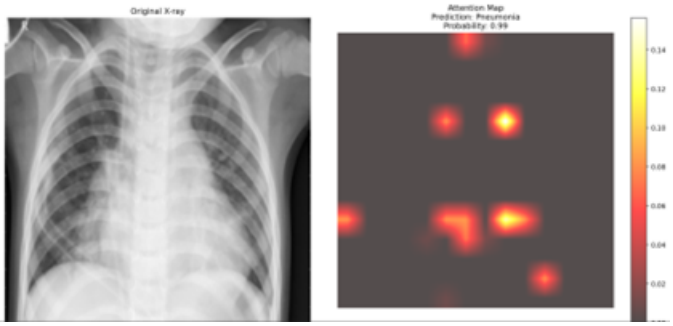
Images as input sequences of patches



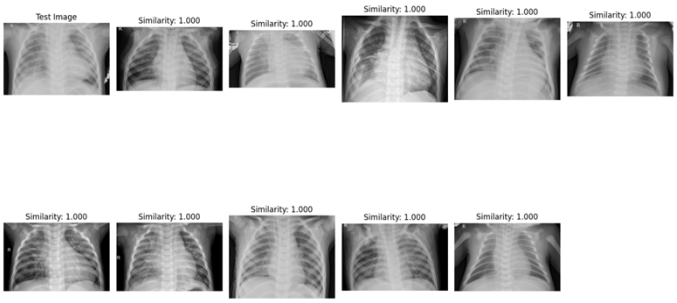
The primary classification output provides a binary prediction, either *normal* or *pneumonia*, with an achieved test accuracy of 81.41%. Beyond binary classification, the model generates meaningful vector representations of each chest X-ray image through its transformer architecture. By transforming input image patches into a high-dimensional embedding space, the model captures complex hierarchical features that encode subtle radiographic characteristics.

The attention mechanism provides the most interpretable output by generating visual heatmaps that highlight regions of the chest X-ray most influential in the model's decision-making process. By averaging the attention weights across multiple attention heads, the model produces a spatial map indicating which image patches contributed most significantly

to the pneumonia classification. These attention maps offer clinically valuable insights, potentially revealing radiographic regions, such as lung consolidation areas or pneumonic infiltrates.



Using the high-dimensional vector representations extracted from the Vision Transformer, we implemented a similarity search algorithm that identifies the top 10 most similar chest X-ray images within the same diagnostic category as the test image. This similarity metric, computed using cosine similarity in the embedding space, allows for nuanced exploration of radiographic pattern similarities.



To enhance the interpretability of these similar images, we employed the *Gemini 1.5 Flash* generative AI model to generate human-readable explanations for each image classification. For each similar image, the language model analyzes the radiographic features and provides a concise, clinically-oriented narrative explaining the key visual characteristics that contribute to its diagnostic classification. The text generated by the language model provides contextual understanding of the model's decision-making process.

By combining advanced computer vision techniques with natural language processing (NLP) models, our methodology presents a comprehensive framework for medical image analysis that goes beyond traditional binary classification, offering a contextual understanding of radiographic diagnostics.

VII. Results Report

The Vision Transformer (ViT) model demonstrated strong performance in detecting pneumonia, achieving an accuracy of 81.41 on the test set. Its classification capabilities were assessed using standard evaluation metrics, providing a detailed understanding of its diagnostic potential. The confusion matrix analysis revealed the model's high precision in distinguishing between normal and pneumonia chest X-rays.

VIII. Key performance metrics include

Overall Test Accuracy: 81.41%. A critical feature of this study was the attention map visualization, which helped provide transparent insights into the model's decision-making process. These heatmaps highlighted the specific areas of the chest X-rays that had the most influence on the classification, offering clinicians a clear view of how the AI reached its conclusions. The attention mechanisms consistently focused on key pneumonia indicators, such as areas of lung consolidation, ground-glass opacities, and changes in tissue density.

The model's ability to retrieve similar images was another key functionality, showing its strength in identifying chest X-rays with similar radiographic features. Using a 768-dimensional embedding space, the model clustered images with comparable diagnostic features, offering a valuable tool for radiological analysis and pattern recognition.

To ensure the accuracy of the insights generated by the model, a context validation process was conducted using semantic similarity analysis. This involved extracting reference material on pneumonia chest X-rays through Wikipedia Loader and using the Hugging Face Embeddings model (sentence-transformers/all-MiniLM-L6-v2) to generate embeddings. The cosine similarity between the reference embeddings and those generated by the model was calculated, and the average similarity score was found to exceed the threshold of 0.5, confirming the relevance of the model's insights. This validation supports the idea that the diagnostic patterns identified by the ViT model align with established medical knowledge.

IX. Conclusion

This research introduces a novel approach to classifying medical images, particularly focusing on the use of Vision Transformers (ViT) for automated pneumonia detection in chest X-rays. By combining state-of-the-art machine learning models with interpretable AI techniques, this study addresses key challenges in medical imaging diagnostics.

X. Key contributions of the study include

- Development of a powerful Vision Transformer model for classifying chest X-rays.

- Implementation of an attention-based mechanism for better model interpretability.

- Use of embedding-based methods to retrieve similar images for advanced analysis.

- Integration of generative AI techniques to explain diagnostic features.

Looking ahead, the research suggests expanding the dataset, investigating multi-class classification for detecting various types of pneumonia, and refining interpretability methods. The proposed solution shows strong potential for enhancing radiological diagnostics, providing valuable support to medical professionals for early disease detection and decision-making.

However, it's important to note that while the model demonstrates impressive results, its purpose is to assist, not replace, clinical expertise. Ongoing validation, interdisciplinary collaboration, and ethical considerations are essential for the development of AI-based diagnostic tools in healthcare.

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